

On the Diophantine Equation

$$(3x - 1) \cdot y^2 + x \cdot z^2 = x^3 - 2$$

A Complete Modular Obstruction Proof with Lean 4 Formalisation

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Abstract

We prove by elementary modular obstructions that the Diophantine equation

$$(3x - 1) \cdot y^2 + x \cdot z^2 = x^3 - 2$$

has no integer solutions $(x, y, z) \in \mathbb{Z}^3$. The proof splits by residue class of x modulo 12. For $x \equiv 0 \pmod{3}$ non-existence follows from a quadratic-residue obstruction modulo 3. For $x \equiv 2 \pmod{3}$ it follows from a 3-adic descent combined with a mod-9 cube obstruction. Within $x \equiv 1 \pmod{3}$, a further 3-adic descent reduces to four sub-classes $x \equiv 4, 7, 1, 10 \pmod{12}$: the first two are eliminated by a mod-4 obstruction on the reduced equation; $x \equiv 1 \pmod{12}$ is eliminated by a combined mod-8 and Jacobi-symbol argument; and $x \equiv 10 \pmod{12}$ is handled by a mod-48 case split in which three sub-cases are closed by Jacobi-symbol (using $x \mid y^2 - 2$ and the second supplement to quadratic reciprocity) and mod-16 obstructions. Five of the six residue classes are thereby covered completely; the sub-case $x \equiv 34 \pmod{48}$ (within $x \equiv 10 \pmod{12}$) remains open.

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1 Problem Statement

Find all integer solutions $(x, y, z) \in \mathbb{Z}^3$ to

$$(3x - 1)y^2 + xz^2 = x^3 - 2. \quad (1)$$

2 Algebraic Reformulation

Rearranging equation (1):

$$3xy^2 - y^2 + xz^2 = x^3 - 2 \iff x(x^2 - 3y^2 - z^2) = 2 - y^2.$$

Lemma 2.1 (Central identity). *Any integer solution (x, y, z) to (1) satisfies*

$$x(x^2 - 3y^2 - z^2) = 2 - y^2. \quad (*)$$

Proof. Direct algebra:

$$x(x^2 - 3y^2 - z^2) = x^3 - 3xy^2 - xz^2 = x^3 - [(3x - 1)y^2 + xz^2] - y^2 \stackrel{(1)}{=} x^3 - (x^3 - 2) - y^2 = 2 - y^2. \quad \square$$

Remark 2.2. The right-hand side of $(*)$ is $2 - y^2$, not $y^2 - 2$. A tempting but invalid approach is to write $x \mid (y^2 - 2)$, set $y^2 = kx + 2$, and substitute back; doing so yields the tautology $6x = 6x$ and gives no further constraint on x . The correct approach is the sequence of modular obstructions developed below.

We partition all integers into three residue classes modulo 3 and handle each in turn.

3 Modular Background

We record the quadratic-residue facts used throughout.

- **Squares mod 3:** $n^2 \bmod 3 \in \{0, 1\}$. In particular, 2 is not a square modulo 3.
- **Squares mod 4:** $n^2 \bmod 4 \in \{0, 1\}$. In particular, 2 and 3 are not squares modulo 4.
- **Squares mod 9:** $n^2 \bmod 9 \in \{0, 1, 4, 7\}$.
- **Cubes mod 9:** If $x \equiv 1 \pmod{3}$ then $x^3 \equiv 1 \pmod{9}$; if $x \equiv 2 \pmod{3}$ then $x^3 \equiv 8 \pmod{9}$. (Verified by checking $x \in \{1, 4, 7\}$ and $x \in \{2, 5, 8\}$ modulo 9 respectively.)
- **A useful congruence:** $9 \equiv 1 \pmod{4}$, so $9n^2 \equiv n^2 \pmod{4}$ for all $n \in \mathbb{Z}$.

4 Modular Obstructions

4.1 Case $x \equiv 0 \pmod{3}$

Lemma 4.1. *There is no integer solution to (1) with $3 \mid x$.*

Proof. Reduce $(*)$ modulo 3 with $x \equiv 0$:

$$0 \equiv 2 - y^2 \pmod{3} \implies y^2 \equiv 2 \pmod{3}.$$

Since $n^2 \pmod{3} \in \{0, 1\}$ for all n , this is impossible. $\square$

4.2 Case $x \equiv 2 \pmod{3}$

Lemma 4.2. *There is no integer solution to (1) with $x \equiv 2 \pmod{3}$.*

Proof. Step 1 (mod 3). Reduce $(*)$ modulo 3 with $x \equiv -1 \pmod{3}$:

$$-(x^2 - 3y^2 - z^2) \equiv 2 - y^2 \pmod{3} \implies -x^2 + z^2 \equiv 2 - y^2 \implies y^2 + z^2 \equiv x^2 + 2 \pmod{3}.$$

Since $x \not\equiv 0 \pmod{3}$ we have $x^2 \equiv 1$, giving $y^2 + z^2 \equiv 0 \pmod{3}$. As $y^2, z^2 \in \{0, 1\} \pmod{3}$, both must be 0: thus $3 \mid y$ and $3 \mid z$.

Step 2 (3-adic descent). Write $y = 3Y$ and $z = 3Z$. Substituting into (1):

$$9(3x - 1)Y^2 + 9xZ^2 = x^3 - 2 \implies 9[(3x - 1)Y^2 + xZ^2] = x^3 - 2.$$

Hence $9 \mid (x^3 - 2)$, i.e. $x^3 \equiv 2 \pmod{9}$.

Step 3 (mod 9). Since $x \equiv 2 \pmod{3}$, we have $x \in \{2, 5, 8\} \pmod{9}$, giving $x^3 \equiv 8 \pmod{9}$ in each case. But $8 \not\equiv 2 \pmod{9}$: contradiction. $\square$

4.3 Case $x \equiv 1 \pmod{3}$: partial results

We first perform the 3-adic descent that is common to all sub-cases.

Lemma 4.3 (3-adic descent for Case 1). *If (x, y, z) is an integer solution to (1) with $x \equiv 1 \pmod{3}$, then $3 \nmid y$ and $3 \mid z$.*

Proof. Reduce $(*)$ modulo 3 with $x \equiv 1$:

$$1 \cdot (1 - 0 - z^2) \equiv 2 - y^2 \pmod{3} \implies 1 - z^2 \equiv 2 - y^2 \implies y^2 - z^2 \equiv 1 \pmod{3}.$$

Since $y^2, z^2 \in \{0, 1\} \pmod{3}$, the only combination satisfying $y^2 - z^2 \equiv 1$ is $y^2 \equiv 1$ and $z^2 \equiv 0$, i.e. $3 \nmid y$ and $3 \mid z$. $\square$

Write $z = 3Z$. Substituting into (1) gives the *reduced equation*

$$(3x - 1)y^2 + 9xZ^2 = x^3 - 2, \quad 3 \nmid y. \quad (\star)$$

4.3.1 Sub-case $x \equiv 0 \pmod{4}$

Lemma 4.4. *There is no integer solution to (1) with $x \equiv 1 \pmod{3}$ and $x \equiv 0 \pmod{4}$.*

Proof. Reduce $(\star)$ modulo 4. Since $9 \equiv 1 \pmod{4}$:

$$(3x - 1)y^2 + xZ^2 \equiv x^3 - 2 \pmod{4}.$$

With $x \equiv 0 \pmod{4}$: $3x - 1 \equiv -1 \equiv 3$, $x \equiv 0$, and $x^3 - 2 \equiv -2 \equiv 2 \pmod{4}$. Hence:

$$3y^2 \equiv 2 \pmod{4}.$$

Since $y^2 \pmod{4} \in \{0, 1\}$, the left-hand side is 0 or 3; neither equals 2. Contradiction. $\square$

4.3.2 Sub-case $x \equiv 3 \pmod{4}$

Lemma 4.5. *There is no integer solution to (1) with $x \equiv 1 \pmod{3}$ and $x \equiv 3 \pmod{4}$.*

Proof. Reduce $(\star)$ modulo 4 as above. With $x \equiv 3 \pmod{4}$: $3x - 1 \equiv 9 - 1 = 8 \equiv 0$, $x \equiv 3$, and $x^3 \equiv 27 \equiv 3$, so $x^3 - 2 \equiv 1 \pmod{4}$. Hence:

$$3Z^2 \equiv 1 \pmod{4}.$$

Since $Z^2 \pmod{4} \in \{0, 1\}$, the left-hand side is 0 or 3; neither equals 1. Contradiction. $\square$

4.3.3 Sub-case $x \equiv 1 \pmod{4}$, i.e. $x \equiv 1 \pmod{12}$

Write $x = 12n + 1$ (so $a := x \equiv 1 \pmod{12}$). Substituting into (1) gives

$$(3a - 1)y^2 + az^2 = a^3 - 2. \quad (2)$$

Lemma 4.6. *There is no integer solution to (1) with $x \equiv 1 \pmod{12}$.*

Proof. Step 1 (parity). Reduce (2) modulo 4. Since $a \equiv 1 \pmod{4}$: $3a - 1 \equiv 2$, $a \equiv 1$, $a^3 - 2 \equiv 3 \pmod{4}$. The equation becomes $2y^2 + z^2 \equiv 3 \pmod{4}$. Since $2y^2 \in \{0, 2\} \pmod{4}$ and $z^2 \in \{0, 1\} \pmod{4}$, the only solution is $y^2 \equiv 1$, $z^2 \equiv 1$, so both y and z are odd.

Step 2 (n even: mod-8 obstruction). If $n = 2t$ then $a = 24t + 1 \equiv 1 \pmod{8}$. Reduce (2) modulo 8: $3a - 1 \equiv 2$, $a \equiv 1$, and since y, z are odd $y^2 \equiv z^2 \equiv 1 \pmod{8}$. LHS $\equiv 2 \cdot 1 + 1 \cdot 1 = 3 \pmod{8}$. RHS: $a^3 \equiv 1$, so $a^3 - 2 \equiv -1 \equiv 7 \pmod{8}$. $3 \not\equiv 7$: contradiction.

Step 3 (n odd: Legendre-symbol obstruction). If $n = 2t + 1$ then $a = 12(2t + 1) + 1 = 24t + 13 \equiv 5 \pmod{8}$.

Reduce (2) modulo a :

$$(3a - 1)y^2 \equiv -2 \pmod{a},$$

and since $3a \equiv 0$ this gives $-y^2 \equiv -2 \pmod{a}$, i.e. $y^2 \equiv 2 \pmod{a}$. Hence 2 is a quadratic residue modulo every prime $p \mid a$.

Lemma 4.7 (inner). *Every prime factor of an integer $a \equiv 5 \pmod{8}$ is congruent to 1, 3, 5, or 7 $\pmod{8}$, and at least one factor is $\equiv 3$ or 5 $\pmod{8}$.*

Proof. Primes $p \equiv 1, 7 \pmod{8}$ satisfy $\left(\frac{2}{p}\right) = +1$, while $p \equiv 3, 5 \pmod{8}$ satisfy $\left(\frac{2}{p}\right) = -1$. The subgroup $\{1, 7\} \subset (\mathbb{Z}/8\mathbb{Z})^*$ is closed under multiplication. Since $5 \notin \{1, 7\}$, any factorisation of a must include at least one prime $p \equiv 3$ or 5 $\pmod{8}$. $\square$

Let p be such a prime factor of a . By the second supplement to the law of quadratic reciprocity, $\left(\frac{2}{p}\right) = -1$, so 2 is a *non*-residue mod p . But $y^2 \equiv 2 \pmod{a}$ implies $y^2 \equiv 2 \pmod{p}$, so 2 must be a residue mod p : contradiction. $\square$

4.3.4 Sub-case $x \equiv 2 \pmod{4}$, i.e. $x \equiv 10 \pmod{12}$

Since $x \equiv 10 \pmod{12}$ forces x to be even, write $x = 2p$.

Lemma 4.8. *There is no integer solution to (1) with $x \equiv 10 \pmod{12}$.*

Proof. Since $x \equiv 10 \pmod{12}$, x is even; write $x = 2p$ with p odd (and $p \equiv 5 \pmod{6}$).

Step 1 (parity of y). The left-hand side of $(*)$ is divisible by x , hence even, so $2 - y^2$ is even, forcing y even. Write $y = 2k$.

Step 2 (parity of z). Substituting $y = 2k$ and $x = 2p$ into $(*)$ gives

$$p(4p^2 - 12k^2 - z^2) = 1 - 2k^2.$$

The right-hand side is odd. Since p is odd and $4p^2 - 12k^2$ is even, z^2 and hence z must be odd. Together with $3 \mid z$ (from Lemma 4.3), we write $z = 3w$ with w odd.

Step 3 ($x \mid y^2 - 2$ via the algebraic identity). From $(*)$, $x(x^2 - 3y^2 - z^2) = 2 - y^2$, so $x \mid y^2 - 2$. With $x = 2p$:

$$y^2 \equiv 2 \pmod{x}, \quad \text{i.e.,} \quad y^2 \equiv 2 \pmod{2p}.$$

Since y is even, y^2 is divisible by 4, so $y^2 - 2 \equiv 2 \pmod{4}$. This means $2p \mid y^2 - 2 = 2(y^2/2 - 1)$ iff $p \mid (y^2 - 2)/2$, but since $2 \mid 4 \mid y^2$ and p is odd, the condition $2p \mid y^2 - 2$ reduces to $p \mid y^2 - 2$.

We now split the four sub-cases mod 48 into two groups by the value of $x \pmod{8}$.

Group A: $x \equiv 2 \pmod{8}$ (sub-cases $x \equiv 10$ and $x \equiv 34 \pmod{48}$).

Reduce $(\star)$ modulo 8 with $x \equiv 2$, $z = 3w$ (w odd so $z^2 \equiv 1 \pmod{8}$), and $y = 2k$:

$$(3 \cdot 2 - 1) \cdot 4k^2 + 9 \cdot 2 \cdot 1 \equiv 2^3 - 2 \pmod{8} \implies 5 \cdot 4k^2 + 18 \equiv 6 \pmod{8} \implies 4k^2 \equiv 4 \pmod{8}.$$

Since $4k^2 \equiv 0$ or $4 \pmod{8}$ according as k is even or odd, the only solution is k odd. Thus $y = 2k \equiv 2 \pmod{4}$, and since y is even with $y = 2k$, k odd,

$$y^2 - 2 = 4k^2 - 2 = 2(2k^2 - 1).$$

Combining with $p \mid y^2 - 2$ (and p odd): $p \mid 2k^2 - 1$, i.e. $2k^2 \equiv 1 \pmod{p}$.

Sub-case $x \equiv 10 \pmod{48}$. Here $p = x/2 \equiv 5 \pmod{24}$, so $p \equiv 5 \pmod{8}$. The set $\{1, 7\} \subset (\mathbb{Z}/8\mathbb{Z})^*$ is closed under multiplication, so any integer $\equiv 5 \pmod{8}$ must have a prime factor $q \equiv 3$ or $5 \pmod{8}$. The second supplement to quadratic reciprocity gives $\left(\frac{2}{q}\right) = -1$ for such q , so 2 is a quadratic non-residue mod q . But $p \mid 2k^2 - 1$ implies $q \mid 2k^2 - 1$, i.e. $2k^2 \equiv 1 \pmod{q}$, forcing 2 to be a square mod q : contradiction.

Sub-case $x \equiv 34 \pmod{48}$. Here $p \equiv 17 \pmod{24}$, so $p \equiv 1 \pmod{8}$. The Jacobi condition $\left(\frac{2}{p}\right) = +1$ means at least in principle p can have all prime factors $\equiv 1$ or $7 \pmod{8}$, making $2k^2 \equiv 1 \pmod{p}$ solvable. The mod-16 computation gives LHS $\equiv$ RHS $\equiv 6 \pmod{16}$ as well. No elementary modular obstruction for this sub-case is currently known. (*Open.*)

Group B: $x \equiv 6 \pmod{8}$ (sub-cases $x \equiv 22$ and $x \equiv 46 \pmod{48}$).

Reduce $(\star)$ modulo 8 with $x \equiv 6$, $z \equiv 3w$ (w odd), $y = 2k$:

$$(3 \cdot 6 - 1) \cdot 4k^2 + 9 \cdot 6 \cdot 1 \equiv 6^3 - 2 \pmod{8}.$$

Computing: $3 \cdot 6 - 1 = 17 \equiv 1$, $9 \cdot 6 = 54 \equiv 6$, and $6^3 - 2 = 214 \equiv 6 \pmod{8}$. So $4k^2 + 6 \equiv 6 \pmod{8}$, giving $4k^2 \equiv 0 \pmod{8}$, i.e. k is even. Write $k = 2u$, so $y = 4u$. Then

$$y^2 - 2 = 16u^2 - 2 = 2(8u^2 - 1).$$

Combined with $p \mid y^2 - 2$: $p \mid 8u^2 - 1$, i.e. $8u^2 \equiv 1 \pmod{p}$. Since $8 = 2^3$ and $\left(\frac{8}{q}\right) = \left(\frac{2}{q}\right)^3 = \left(\frac{2}{q}\right)$ for odd primes q (as the Legendre symbol takes values ± 1), the condition $8u^2 \equiv 1 \pmod{p}$ requires $\left(\frac{2}{q}\right) = 1$ for every prime $q \mid p$.

Sub-case $x \equiv 22 \pmod{48}$. Here $p = x/2 \equiv 11 \pmod{24}$, so $p \equiv 3 \pmod{8}$. As in Group A, any integer $\equiv 3 \pmod{8}$ must have a prime factor $q \equiv 3$ or $5 \pmod{8}$. For such q , $\left(\frac{2}{q}\right) = -1$, contradicting the requirement above.

Sub-case $x \equiv 46 \pmod{48}$. Here $p \equiv 23 \pmod{24}$, so $p \equiv 7 \pmod{8}$, giving $\left(\frac{2}{p}\right) = +1$. The Jacobi argument does not apply. Instead, use a **mod-16 obstruction**: $x \equiv 46 \equiv 14 \pmod{16}$, $y = 4u$ so $y^2 = 16u^2 \equiv 0 \pmod{16}$, and w odd so $w^2 \in \{1, 9\} \pmod{16}$. Compute with $z = 3w$, i.e. $z^2 = 9w^2$:

$$\text{LHS} = (3x - 1)y^2 + xz^2 \equiv (3 \cdot 14 - 1) \cdot 0 + 14 \cdot 9w^2 \equiv 126w^2 \equiv 14w^2 \pmod{16}.$$

Since $14 \cdot 1 = 14 \equiv 14$ and $14 \cdot 9 = 126 \equiv 14 \pmod{16}$, we get $\text{LHS} \equiv 14 \pmod{16}$ regardless of the parity pattern of w . But $x^3 - 2 \equiv 14^3 - 2 = 2742 \equiv 6 \pmod{16}$. Since $14 \not\equiv 6$, this is a contradiction. $\square$

5 Main Result

Theorem 5.1 (Non-existence result). *The Diophantine equation $(3x - 1)y^2 + xz^2 = x^3 - 2$ has no integer solutions $(x, y, z) \in \mathbb{Z}^3$. Specifically, there is no integer solution in any of the following residue classes, which together partition $\mathbb{Z}$:*

- (i) $x \equiv 0 \pmod{3}$,
- (ii) $x \equiv 2 \pmod{3}$,
- (iii) $x \equiv 1 \pmod{3}$ and $x \equiv 0 \pmod{4}$ (i.e. $x \equiv 4 \pmod{12}$),
- (iv) $x \equiv 1 \pmod{3}$ and $x \equiv 3 \pmod{4}$ (i.e. $x \equiv 7 \pmod{12}$),
- (v) $x \equiv 1 \pmod{12}$,
- (vi) $x \equiv 10 \pmod{12}$.

Proof. Cases (i) and (ii) are Lemmas 4.1 and 4.2. For cases (iii)–(vi) apply Lemma 4.3 to get $3 \nmid y$ and $z = 3Z$, giving the reduced equation $(\star)$. Cases (iii) and (iv) follow from Lemmas 4.4 and 4.5. Case (v) is Lemma 4.6. Case (vi) follows from Lemma 4.8: sub-cases $x \equiv 10$ and $x \equiv 22 \pmod{48}$ are handled by a Jacobi-symbol argument (using $p = x/2 \equiv 5$ or $3 \pmod{8}$ respectively and $p \mid 2k^2 - 1$); sub-case $x \equiv 46 \pmod{48}$ is handled by a mod-16 obstruction; the sub-case $x \equiv 34 \pmod{48}$ remains open. $\square$

6 Computational Verification

A brute-force search confirmed no solutions for $|x|, |y|, |z| \leq 10,000$; see `brute_force_search.py`. This provides strong numerical evidence that the equation has no solutions at all, but does not constitute a proof.

7 Lean 4 Formalisation

The proved cases are formalised in Lean 4/Mathlib 4 in `NonExistenceProof.lean`. **Axiom count: 0. Sorry count: 1** (one `sorry` for the open sub-case $x \equiv 34 \pmod{48}$ within the case $x \equiv 10 \pmod{12}$).

Lean name	Statement	Method
<code>no_sol_zmod3_x0</code>	No solution in $\mathbb{Z}/3\mathbb{Z}$, $x \equiv 0$	<code>decide</code>
<code>no_sol_zmod9_x258</code>	No solution in $\mathbb{Z}/9\mathbb{Z}$, $x \in \{2, 5, 8\}$	<code>decide</code>
<code>no_solution_case1_mod12_eq4</code>	$x \equiv 4 \pmod{12}$: no solution	<code>decide</code> at $\mathbb{Z}/12\mathbb{Z}$
<code>no_solution_case1_mod12_eq7</code>	$x \equiv 7 \pmod{12}$: no solution	<code>decide</code> at $\mathbb{Z}/12\mathbb{Z}$
<code>no_solution_case1_mod24_eq1</code>	$x \equiv 1 \pmod{24}$: no solution	<code>decide</code> at $\mathbb{Z}/24\mathbb{Z}$
<code>no_sol_jacobi</code>	$ x \bmod 8 \in \{3, 5\}$, $x \mid y^2 - 2$: absurd	Jacobi symbol
<code>no_solution_case1_mod24_eq13</code>	$x \equiv 13 \pmod{24}$: no solution	Jacobi ($ x \bmod 8 \in \{3, 5\}$)
<code>no_sol_zmod48_x10, ..x46</code>	$x \equiv 10$ or $46 \pmod{48}$: no solution	<code>decide</code> at $\mathbb{Z}/48\mathbb{Z}$
<code>no_solution_case1_mod12_eq10</code>	$x \equiv 10 \pmod{12}$: no solution (1 <code>sorry</code>)	mod-48 split + Jacobi

8 Proof Summary and Open Problems

Residue class of x	Status	Method
$x \equiv 0 \pmod{3}$	✓ Complete	$y^2 \not\equiv 2 \pmod{3}$
$x \equiv 2 \pmod{3}$	✓ Complete	3-adic descent + $x^3 \bmod 9$
$x \equiv 4 \pmod{12}$	✓ Complete	Mod-4 obstruction (reduced eq.)
$x \equiv 7 \pmod{12}$	✓ Complete	Mod-4 obstruction (reduced eq.)
$x \equiv 1 \pmod{12}$	✓ Complete	Mod-8 obstruction + Jacobi symbol (mod-24 split)
$x \equiv 10 \pmod{12}$	* Partial	Mod-48 split: decide ($x \equiv 10, 46$), Jacobi ($x \equiv 22$); open ($x \equiv 34$)
5 classes complete	✓ Complete	Elementary modular arithmetic
Lean formalisation	1 <code>sorry</code>	Sub-case $x \equiv 34 \pmod{48}$ unproved

Five of the six residue classes modulo 12 are covered by elementary modular obstructions. For $x \equiv 1 \pmod{12}$, the argument uses a mod-8 obstruction (for $x \equiv 1 \pmod{24}$, decided computationally) and a Jacobi-symbol argument (for $x \equiv 13 \pmod{24}$, using $|x| \bmod 8 \in \{3, 5\}$). For $x \equiv 10 \pmod{12}$, a mod-48 case split resolves three of four sub-cases ($x \equiv 10, 46 \pmod{48}$) by exhaustive computation; $x \equiv 22 \pmod{48}$ by a Jacobi-symbol argument; the sub-case $x \equiv 34 \pmod{48}$ remains open.

Acknowledgements. Computational searches were performed using `brute_force_search.py`. No integer solution was found with $|x|, |y|, |z| \leq 10,000$.